

Technology Stack

Date	1 July 20206
Team ID	SWTID-2026-4523
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Technology Stack:

The technology stack defines the collection of programming languages, frameworks, libraries, databases, and development tools used to design, develop, and deploy the Credit Card Approval Prediction System. It provides a strong foundation for building a secure, scalable, and efficient web application. The project uses HTML, CSS, and JavaScript to create an interactive user interface, while Python and the Flask framework handle server-side processing and communication with the machine learning model. Scikit-learn is used to train and evaluate the prediction model, and Pandas and NumPy are used for data preprocessing and analysis. The trained model is stored using Pickle for efficient loading during prediction. GitHub is used for version control and project management, ensuring effective collaboration and code maintenance.

Technology Stack Details:

S.NO	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript	Used to create a responsive and user-friendly interface for entering applicant details and displaying prediction results.
2	Backend / Server-Side	Python, Flask	Flask handles HTTP requests, processes user input, loads the trained machine learning model, and returns prediction results efficiently.
3	Backend / Server-Side	CSV Dataset (Application Record & Credit Record)	Used to store and preprocess applicant information for training and testing the machine learning model.
4	Cloud / Hosting / Deployment	Localhost (Flask Development Server)	Used for developing, testing, and demonstrating the application before deployment.

5	Version Control & CI/CD	GitHub	Used to store the project, manage versions, and collaborate with team members.
6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy, Pickle	Scikit-learn is used for model training and prediction, Pandas and NumPy for data preprocessing, and Pickle for saving and loading the trained model.